
Snap-HyRE: When Generated Legal Queries Help Legal RAG

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Abstract

Legal RAG often fails before the answer model sees useful law: the question and the corpus speak different languages. A bar-exam question describes “a driver and a banana peel”; the supporting passage states a rule about proximate cause. HyDE addresses this by letting the model write a synthetic passage in the corpus’s voice and searching with that text instead of the original question (Gao et al., 2023). We introduce Snap-HyRE, a two-call extension that mirrors expert legal reasoning. The first call writes a private draft answer (an intuition about the controlling doctrine) and, separately, a legal search passage; only the passage is used for retrieval, and the draft is withheld from the second call so the final answer is judged against the corpus, not the guess. On BarExamQA, Snap-HyRE improves raw-question RAG by up to 5.1 percentage points across three model sizes, raises top-five evidence retrieval from 1.4% to 9.5–12.1%, and matches or exceeds the labeled gold-passage oracle on Gemma 4 26B and Llama 3.3 70B. On HousingQA, where the question already shares vocabulary with the target statute, Snap-HyRE holds answer parity with raw-question RAG, and a single in-prompt statute example then lifts retrieval above the raw baseline. Pooled across five model–dataset configurations, Snap-HyRE averages +1.1pp over raw-question RAG and is at parity with HyDE. The pattern is clean: Snap-HyRE pays off in proportion to the question–corpus vocabulary gap.

1. Introduction

Legal RAG should work well for legal question answering: many answers turn on statutes, regulations, court holdings, or compact doctrinal rules, and retrieval can put that text in front of a language model (Lewis et al., 2020). The hard part is often the first query. A bar-exam fact pattern may mention a driver, crutches, and a banana peel, while the support passage states a well-established rule about proximate cause. A housing question may name a state, while a national corpus that mixes statutes from every state can return a plausible-sounding rule from the wrong jurisdiction. Final answer accuracy alone hides these failures.

A common fix is to let the model first write the kind of passage it expects to find and search with that text instead of the raw question. HyDE asks the model for a short hallucinated authority — a synthetic passage in the corpus’s voice — and embeds that text for retrieval (Gao et al., 2023). Query expansion and rewriting use related generated search text (Wang et al., 2023). The idea is attractive in law because a model can often name the legal category behind a fact pattern before retrieval sees the corpus. The same methods can also fail by abstracting away anchors, choosing the wrong legal frame, or retrieving plausible but irrelevant authority.

We introduce Snap-HyRE, a fixed two-call generated-query method that mirrors expert legal reasoning. A practitioner facing a fact pattern forms a quick intuition about what is at issue, looks up the controlling authority by that intuition, and then judges the evidence on its own merits rather than anchoring on the first guess. Snap-HyRE encodes the same loop. The first model call writes two objects: a private draft answer (the “snap” intuition) and a separate legal search passage in the style of the authority that would resolve the question. Only the search passage is embedded for

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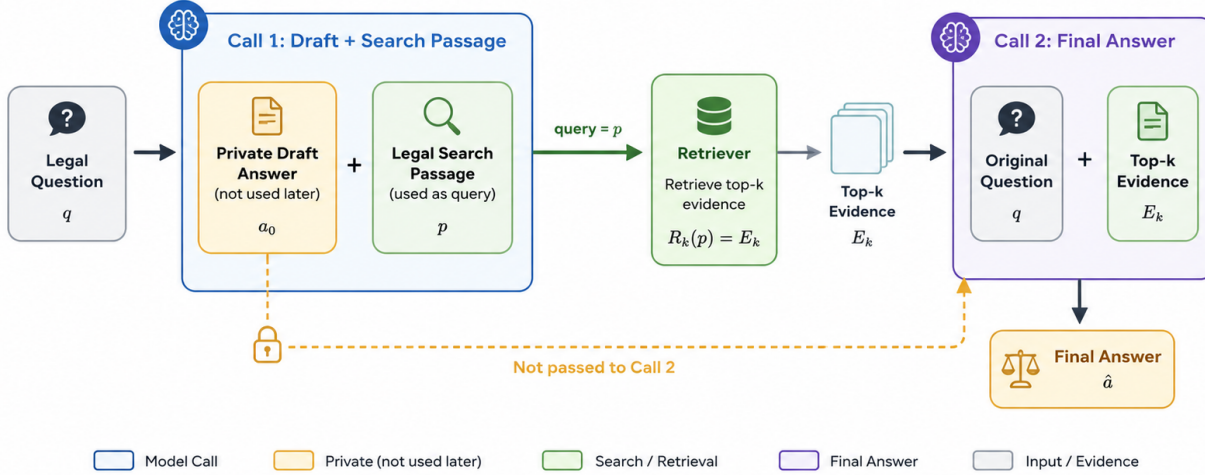


Figure 1. Snap-HyRE: the first model call writes a private draft answer and a separate legal search passage; only the search passage is used to retrieve evidence. The second model call sees only the original question and the top- k retrieved passages, and produces the final answer. The draft is never passed to the second call.

retrieval; the draft is not handed forward. The second call answers the original question from the retrieved legal text, judging the evidence without anchoring on its own first guess. The name reflects this structure: “snap” is the quick private draft answer, and HyRE abbreviates Hypothetical Reasoning Embeddings, the legal passage embedded for retrieval.

We report a two-benchmark study chosen for opposite ends of the question-corpus vocabulary gap. BarExamQA is the weak-query regime: fact-pattern questions describe accidents, transactions, and procedures in narrative prose, while the support corpus uses doctrinal vocabulary (“proximate cause,” “hearsay exception,” “adverse possession”). The raw question and the relevant rule passage barely share words. HousingQA is the strong-query regime: statutory-entailment questions ask directly about landlord and tenant terms (“Can a landlord evict a tenant for committing waste?”), so the question vocabulary already overlaps with the housing statutes that contain the answer.

Snap-HyRE helps in proportion to that gap. On BarExamQA it improves raw-question RAG by +2.4pp, +4.0pp, and +5.1pp for Llama 3.1 8B, Gemma 4 26B, and Llama 3.3 70B; it raises BarExamQA Hit@5

from 1.4% to 9.5–12.1% for the three generators; and it matches or exceeds the labeled gold passage on the two larger models. On HousingQA, where raw queries are already corpus-shaped, Snap-HyRE delivers answer parity rather than lift. Pooled across all five signed model-dataset cells, Snap-HyRE averages +1.1pp over raw-question RAG and is at parity with HyDE.

We define Snap-HyRE as a fixed two-call legal RAG method, compare it against raw-question RAG, HyDE, and oracle evidence controls on BarExamQA and HousingQA, and report when the method helps by tracking both answer accuracy and whether gold evidence appears in the top five retrieved passages. Structured query rewriting and additional diagnostics are in the appendix.

2. Background and Related Work

Generated search text. RAG combines a generator with an evidence store: a searchable corpus of text that the system retrieves from before answering (Lewis et al., 2020). Its legal promise is simple: retrieve the relevant rule or statute, then answer from it. But the query and the corpus often use different language. HyDE addresses this by generating a hypothetical document and embedding that text for retrieval (Gao

et al., 2023). Query2doc and related expansion methods similarly use a language model to turn sparse query text into richer retrieval handles (Wang et al., 2023). For legal tasks, the useful generated text is usually not a longer version of the question. It is a passage in the style of the authority the retriever should find.

Legal RAG evaluation. Legal evaluation benchmarks and corpora have standardized many legal-understanding tasks (Guha et al., 2023; Chalkidis et al., 2022; Henderson et al., 2022), and recent work has emphasized the risk of unsupported legal assistant answers (Magesh et al., 2024). Legal RAG benchmarks such as LegalBench-RAG and Legal RAG Bench argue for evaluating retrieval and answering together (Pipitone & Alami, 2024; Butler & Butler, 2026). We follow that discipline but keep the methods fixed: every row uses a predetermined retrieval interface, and the analysis asks whether failure comes from missing evidence or from using retrieved evidence poorly.

3. Snap-HyRE

3.1. Design Rationale

Snap-HyRE mirrors a small piece of how an expert reads a legal scenario. A practitioner facing a fact pattern forms a quick intuition about what is at issue (proximate cause? a hearsay exception? a state-specific notice rule?), looks up the controlling authority by that intuition, and then judges the evidence on its own merits rather than anchoring on the initial guess. Snap-HyRE encodes the same loop. The first call is the intuition step: it produces a draft answer and a separate legal search passage written in the style of the controlling authority. The second call is the evidence step: it sees only the original question and the retrieved evidence, never the draft, so the final answer is judged against what the corpus actually says. Figure 1 shows the resulting two-call pipeline.

3.2. Two Calls, One Retrieval Query

Snap-HyRE separates query formation from final answering. The same model G_θ is used for both calls; only the prompt template differs (the superscript). Given a formatted legal question q , the first call produces a private draft answer a_0 and a legal search passage p :

$$(a_0, p) = G_\theta^{\text{snap}}(q). \quad (1)$$

The retriever R_k embeds only p and returns the top- k retrieved passages $E_k = R_k(p)$. The final answer is

then

$$\hat{a} = G_\theta^{\text{answer}}(q, E_k), \quad (2)$$

with a_0 withheld from the final answer context. The first call can use the model’s prior knowledge of law to choose vocabulary; the second call must answer from the retrieved corpus text.

Each Snap-HyRE row follows three steps: validate that the first call produced both a formatted draft answer and a clean search passage; retrieve and rerank using only the search passage; answer the original question from the retrieved passages without exposing the draft answer. The retrieval passage is constrained to be short and reference-like: it should describe the controlling rule, statutory condition, legal relationship, or doctrinal test without answer labels or an Answer: header.

The difference from HyDE is narrow but consequential. HyDE writes a legal passage directly from the question; Snap-HyRE first forms a tentative legal answer and uses that private judgment only to decide what kind of passage to search with. For a torts fact pattern about a broken leg, crutches, and a later fall, this turns an accident-heavy query into a passage about proximate cause and foreseeable subsequent injury (Figure 2).

3.3. Why It Works: Two Guardrails

The two-call structure gives the method two guardrails that follow from the formulation above. First, the draft a_0 is private: it can shape what to look for, but it is not handed in as evidence, so the second call avoids the anchoring bias of treating its own first guess as a fact. Second, the search passage p is written like legal authority rather than chain-of-thought reasoning: it names the rule or statutory condition that would resolve the question, not the steps the model used to get there. The intuition step is allowed to use model priors; the evidence step is forced back to the corpus.

4. Evaluation

4.1. Two Legal Retrieval Regimes

BarExamQA and HousingQA sit at opposite ends of the question-corpus vocabulary gap. BarExamQA questions are fact patterns in narrative prose, while the support corpus uses doctrinal vocabulary — a large gap that a generated legal passage can close. HousingQA questions are statutory-entailment questions framed in landlord-tenant terms, sharing vocabulary with the housing statutes that answer them — a small

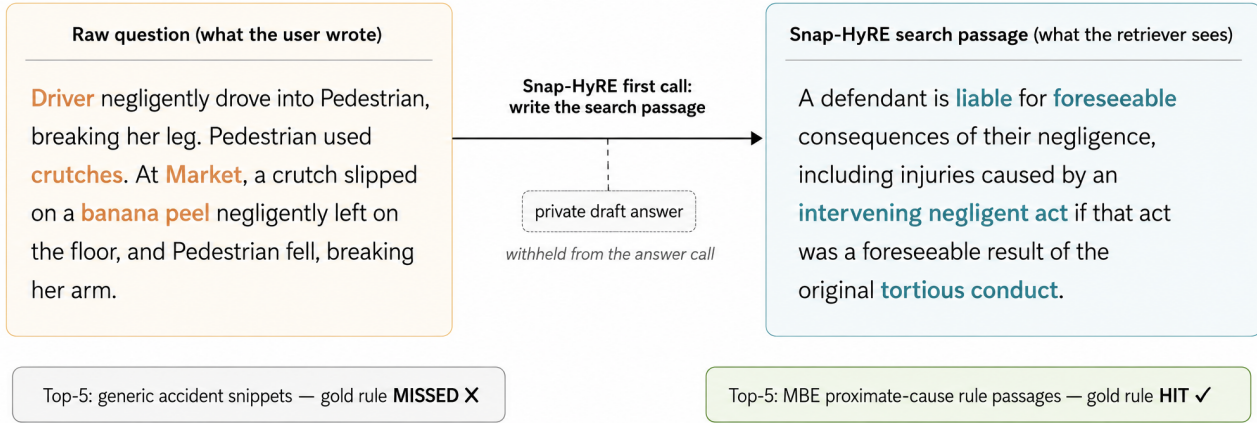


Figure 2. What the two methods send to the retriever for one BarExamQA item (mbe_1175, Gemma 4 26B). Raw-question RAG embeds the narrative fact pattern; Snap-HyRE embeds a short doctrinal passage about foreseeable intervening injury, which is the language of the rule the retriever actually needs to surface.

gap where raw queries are already corpus-shaped. The main paper reports both benchmarks, exact-scored.

BarExamQA. BarExamQA contains 1,195 multiple-choice bar-exam questions in the evaluation split used here. The retrieval corpus is a frozen support-passage collection with official gold passage ids. Hit@5 and MRR@5 therefore measure whether the retriever exposes the labeled rule passage before the answer model chooses an option.

HousingQA. HousingQA contains 6,853 yes/no statutory entailment questions over state housing statutes. The corpus contains many jurisdictions, and each question names the relevant state. All main HousingQA retrieval rows apply that state as a metadata filter before ranking. Rows that do not use this filter are excluded from the main HousingQA comparison.

4.2. Models, Baselines, and Retrieval Stack

We evaluate three model configurations: Llama 3.1 8B Instant, Gemma 4 26B, and Llama 3.3 70B Versatile. Within each row the same model serves as both the query generator (when one is needed) and the final answer model, so accuracy lift cannot be attributed to a stronger reader.

We compare four deployable retrieval methods and one oracle control. All non-oracle rows answer from $k = 5$ retrieved passages.

- LLM only answers the original question without retrieval.
- Raw question RAG retrieves with the question text,

then answers from the question plus top- k evidence.

- HyDE retrieves with a generated legal passage, without a draft answer prior.
- Snap-HyRE (ours) retrieves with a draft-guided legal search passage; the draft answer is private and is not shown to the final answer call.
- Gold passage supplies the labeled passage directly as an oracle control for evidence use (not a deployable retrieval method).

Structured rewrite and gold-plus-neighbor diagnostics appear in the appendix; route labels are retained in the released artifacts for reproducibility.

Retrieval stack. Runs use seed 42, temperature 0, and a 2048-token answer cap. If the model produces a malformed generated passage or no parseable final answer, the row fails rather than silently falling back to raw-question retrieval. Retrieval uses Chroma collections built with Alibaba-NLP/gte-large-en-v1.5 embeddings; a cross-encoder (ms-marco-MiniLM-L-6-v2) then reranks the candidate set before the top- k cut.

Generated-query evaluation. Generated-query rows are evaluated in two stages. First, the query-generation call writes the HyDE passage or Snap-HyRE search passage. Then retrieval stores the ranked document ids. The answer run replays those ids strictly; an invalid generated passage or missing cache entry is not replaced with raw-question retrieval. This keeps the comparison about the method’s actual search text.

Per-dataset retrieval interfaces. BarExamQA retrieval uses the shared fact pattern and question stem; generated methods may see option text but no correctness marker. HousingQA retrieval applies the state metadata filter described above. The filter is not a Snap-HyRE-specific feature: it is the benchmark interface used by every HousingQA row in the main paper.

4.3. Answer and Evidence Metrics

The downstream metric is exact answer accuracy. BarExamQA uses the last valid multiple-choice answer line, and HousingQA uses Answer: Yes or Answer: No. Retrieval is measured by Hit@5 and MRR@5: whether any labeled support passage appears in the top five, and how early it appears.

We also report efficiency as the number of model calls per question, the tokens the answer call processes per question (the “answer-stage tokens”), and correct answers per million answer-stage tokens. Two-call methods are not penalized in this count: the first-stage query-generation call is counted as part of the method footprint and logged separately. The answer-stage token count shows how much context the final answer call sees once retrieval has selected evidence.

The main tables include only validated full rows. Missing cells are not imputed. Appendix A.1 lists the inclusion rule, coverage grid, supplemental controls, top- k diagnostics, efficiency table, and reproducibility details.

5. Results

5.1. When Snap-HyRE Helps Answers

Snap-HyRE helps in proportion to the question-corpus vocabulary gap. BarExamQA fact patterns describe accidents and transactions in narrative prose while the support corpus uses doctrinal language; raw-question RAG retrieves the labeled rule passage in only 1.4% of cases at Hit@5. Snap-HyRE raises Hit@5 to 9.5%, 12.1%, and 11.0% across the 8B, 26B, and 70B models, and those retrieval gains convert to answer gains over raw-question RAG: 54.5% to 56.9% for Llama 3.1 8B, 78.0% to 82.0% for Gemma 4 26B, and 74.6% to 79.7% for Llama 3.3 70B. Averaged across the three models, Snap-HyRE reaches 72.9%, compared with 69.0% for raw-question RAG and 72.2% for HyDE.

HousingQA is the opposite end of that axis. Statutory-entailment questions ask directly about landlord and tenant terms (“Can a landlord evict a tenant for committing waste?”), so the question already shares vocabulary with the target housing statutes; raw-question

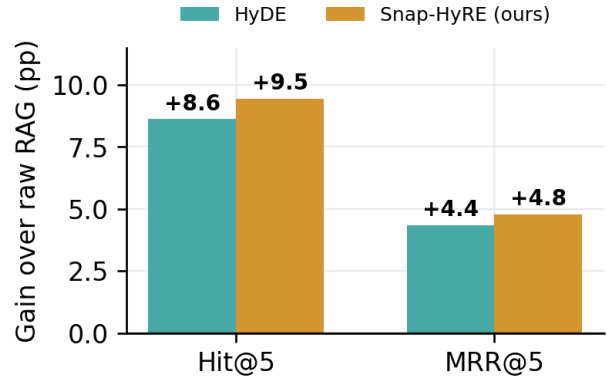


Figure 3. Mean BarExamQA retrieval gains over raw-question RAG. Generated search passages improve both top-five exposure and rank.

retrieval reaches 36.9% Hit@5 with no query generation. With less of a gap to close, Snap-HyRE delivers answer parity rather than lift: 59.0% / 59.6% on 8B / 70B versus raw-question RAG at 62.3% / 62.1%. Pooled across all five evaluated model-dataset configurations, Snap-HyRE averages 67.4% accuracy versus 66.3% for raw-question RAG (+1.1pp) and 67.6% for HyDE (parity). §5.7 shows that even the small HousingQA gap can be closed: a single statute-style exemplar in the Snap-HyRE prompt recovers Hit@5 above the raw baseline.

5.2. Did the Right Evidence Appear?

Table 2 separates retrieval from answer selection, and Figure 3 plots the BarExamQA gains. On BarExamQA, generated legal passages improve exact evidence exposure: raw retrieval averages 1.4% Hit@5, HyDE averages 10.0%, and Snap-HyRE averages 10.9%. MRR@5 moves in the same direction, from 0.7% for raw-question retrieval to 5.0% for HyDE and 5.4% for Snap-HyRE.

On HousingQA the picture is reversed because raw queries already share vocabulary with the corpus. Raw-question retrieval reaches 36.9% Hit@5, and the generated-query rows trade some labeled-evidence exposure for the broader corpus context they retrieve (still useful for the answer model, as the parity accuracy in Table 1 shows). Appendix A.3 reports Hit@1/3/5/10 and MRR@10 from full top-10 retrieval runs.

The extra evidence is useful but incomplete. Snap-HyRE converts a +9.5pp Hit@5 and +4.8pp MRR@5 retrieval shift into +3.9pp average BarExamQA answer accuracy over raw-question RAG, while exact gold evidence still appears in only about one in nine top-five lists. The remaining gap points to ranking,

Table 1. Main answer accuracy (%). Bold marks the largest non-oracle value in each column. Averages are descriptive over available cells.

Method	BarExamQA				HousingQA				Pooled
	8B	26B	70B	Avg.	8B	26B	70B	Avg.	
LLM	57.3	80.8	78.7	72.3	55.4	—	44.8	50.1	63.4
Raw question RAG	54.5	78.0	74.6	69.0	62.3	66.1	62.1	63.5	66.3
HyDE	56.1	80.3	80.2	72.2	59.1	—	62.2	60.7	67.6
Snap-HyRE (ours)	56.9	82.0	79.7	72.9	59.0	—	59.6	59.3	67.4
Gold	60.0	78.6	79.2	72.6	64.3	—	67.3	65.8	69.9

Table 2. Evidence exposure at $k = 5$ (%). Bold marks the largest value in each column. Generated-query rows average available model-specific generators; HousingQA uses state-filtered retrieval.

Method	BarExamQA		HousingQA	
	Hit@5	MRR@5	Hit@5	MRR@5
Raw question RAG	1.4	0.7	36.9	23.3
HyDE	10.0	5.0	30.9	19.7
Snap-HyRE (ours)	10.9	5.4	26.3	16.6

option mapping, and evidence use.

5.3. How the Win Happens

Table 3. Worked BarExamQA example (mbe_1175, Gemma 4 26B). Gold answer: B (driver liable for both injuries).

Method	Pred.	Top-5 retrieval
Raw question RAG	D ✗	Generic accident and intervening-cause material; gold rule passage missed.
HyDE	D ✗	Drifts toward proximate cause but stays broad; gold rule passage missed.
Snap-HyRE (ours)	B ✓	Anchored on foreseeable subsequent injury; surfaces MBE proximate-cause rule passages.

On this item, raw-question RAG retrieves generic accident material and HyDE drifts toward proximate cause without landing on the rule; Snap-HyRE writes a short passage about foreseeable intervening injury, surfaces the MBE proximate-cause rule, and picks (B). The win comes from the search query, not from passing the answer through: the second call sees only the

original question and the retrieved legal text, never the private draft.

5.4. Snap-HyRE Matches the Gold Oracle on Larger Models

On BarExamQA, Snap-HyRE matches or exceeds the labeled gold passage on the two larger models. Gemma 4 26B reaches 82.0% with Snap-HyRE versus 78.6% with the gold passage; Llama 3.3 70B reaches 79.7% versus 79.2%. Only Llama 3.1 8B trails the oracle (56.9% versus 60.0%). Averaged across the three model slices, Snap-HyRE reaches 72.9% and the gold-passage oracle reaches 72.6%. The retrieved top-5 list does more work than the labeled rule alone: surrounding legal passages supply context the answer model uses. The same pattern shows up at the oracle level in Appendix A.2: adding neighbors to the gold passage also improves accuracy on Llama 3.1 8B and Gemma 4 26B.

5.5. Gains Grow with Model Size

Snap-HyRE gains over raw-question RAG grow with model size on BarExamQA: +2.4pp at 8B, +4.0pp at 26B, +5.1pp at 70B. Hit@5 also grows from 9.5% (8B) to 12.1% (26B) and 11.0% (70B). Bigger models both write more useful search text and convert the retrieved evidence into more answers.

5.6. Best Answer-Token Efficiency Among Non-Oracle Methods

The appendix efficiency table (Table 8) reports correct answers per million tokens that the answer call processes. Snap-HyRE answers 268.3 questions correctly per million answer-stage tokens, compared with 244.5 for HyDE and 241.5 for raw-question RAG. The first-stage query-generation call is logged separately and is not included in this denominator; the lift comes from a shorter average answer context (2,377 vs 2,634 tokens for raw-question RAG) at higher accuracy. The gold-passage control reaches 404.5 correct per million tokens because it sees only the single labeled passage.

5.7. Sharpening the Intuition with One Example

The first call’s prompt can be improved with a single example. We add one real passage from the target corpus to the Snap-HyRE generation prompt — its question, answer, and document id stripped — and re-run the first call. On 500-question probe slices for Gemma 4 26B (Table 11), this lifts retrieval on every benchmark we tested. The size of the lift tracks how unfamiliar the target corpus phrasing is: small on BarExamQA (+0.6pp Hit@5), where Snap-HyRE already writes in doctrinal language; +3.0pp on state-filtered HousingQA, where statutes use specific terms of art; and +7.6pp on Legal-Link-EU, where EU citation forms carry the bulk of the relevance signal. Paired answer accuracy on the HousingQA 500-question slice is 62.8% with the example versus 63.0% for plain Snap-HyRE (-0.20pp , $p = 1.0$): the example recovers retrieval without costing answer accuracy. The same expert-reasoning loop runs, with the first-call intuition calibrated by one corpus example; on an Ohio writ-of-execution question, for instance, the example-prompted passage names “writ of restitution” and “execution” in statute style and retrieves the gold Ohio statute at rank 3, where plain Snap-HyRE drifts to neighboring writ-of-possession language.

Legal-Link-EU also marks where Snap-HyRE’s advantage ends. Its questions cite source and target document anchors directly, so the raw question already is a strong corpus search (Hit@5 90.0% vs plain Snap-HyRE 68.2%). When the user has already framed their need as a search query, the intuition step isn’t needed. The typical legal scenario in practice is the other way around — a fact pattern, a tenant complaint, an audit notice that the user has to turn into a search query before any retrieval can happen — and that is where Snap-HyRE targets its gains. The one-example variant still helps in the boundary case: it lifts Legal-Link-EU Hit@5 from 68.2% to 75.8%.

6. Conclusion

Snap-HyRE encodes a small piece of expert legal reasoning into a fixed two-call pipeline: form an intuition about the controlling doctrine, retrieve evidence by that frame, then judge the evidence on its own without anchoring to the intuition. The lift over raw-question RAG tracks the question-corpus vocabulary gap because that gap is precisely the regime where the intuition step matters — when the question’s surface words don’t yet point to the right authority. On BarExamQA, where fact-pattern questions barely share words with the doctrinal corpus, Snap-HyRE improves Hit@5 and final accuracy across all three model sizes, matches or exceeds the labeled gold-passage oracle on Gemma 4 26B and Llama 3.3 70B, and delivers the best correct-answer count per million answer-stage tokens among non-oracle methods. On HousingQA, where statutory-entailment questions already share vocabulary with the housing statutes, Snap-HyRE reaches answer parity with raw-question RAG; adding one real statute as an example in the first-call prompt then raises Hit@5 above the raw baseline on a 500-question slice while holding answer accuracy at parity. Pooled across all five evaluated model-dataset configurations, Snap-HyRE averages +1.1pp over raw-question RAG and is at parity with HyDE.

The takeaway is practical: when the raw question is a poor retrieval handle, let the model form an intuition before it searches, but force the final answer back to the corpus. Snap-HyRE is the right tool in that regime and a clean parity baseline when the question already reads like a passage from the corpus.

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A. Appendix

A.1. Result Inclusion Policy

A row is used as a paper claim only when the result record names the dataset, provider, method, answer depth, metric values, and any row note needed to interpret the result. Rows without validated full results are omitted from the main comparison rather than imputed.

- Main benchmarks: BarExamQA and HousingQA.
- Route labels: groq-llama8b, or-gemma4-26b, groq-llama70b.
- Main modes: llm_only, rag_simple, rag_hyde, snap_hyre, and golden_passage. rag_rewrite is a supplemental generated-query control, and golden_plus_neighbors is retained only as a neighbor-dilution diagnostic.
- Answer depth: $k = 5$ for retrieval rows.
- HousingQA interface: retrieval rows must use the jurisdiction state filter to count as main-interface rows.

The main matrix has 31 signed cells out of 42 expected (BarExamQA 21/21; HousingQA 10/21).

A.2. Supplemental Controls

The supplemental tables separate the fixed-method comparison from controls that are useful for interpreting failure modes: structured rewrites, oracle context, neighbor dilution, and answer-stage token accounting.

Table 4. Dataset-level and pooled deltas. Mean percentage-point differences across signed model slices.

Slice	Snap vs. raw pp	HyDE vs. Snap pp	Gold vs. raw pp
BarExamQA	+3.8 (n=3)	-0.7 (n=3)	+3.6 (n=3)
HousingQA	-2.9 (n=2)	+1.4 (n=2)	+3.6 (n=2)
Pooled	+1.1 (n=5)	+0.1 (n=5)	+3.6 (n=5)

Table 5. Snap-HyRE against non-oracle controls. Cells are accuracy percentages; bold marks the strongest non-oracle row per slice.

Dataset	Model	LLM	Raw question RAG	HyDE	Snap-HyRE	Rewrite	Snap vs. raw pp	Best main non-oracle
BarExamQA	Llama 3.1 8B	57.3	54.5	56.1	56.9	57.3	+2.4	LLM (57.3)
BarExamQA	Gemma 4 26B	80.8	78.0	80.3	82.0	80.7	+4.0	Snap-HyRE (ours) (82.0)
BarExamQA	Llama 3.3 70B	78.7	74.6	80.2	79.7	77.2	+5.1	HyDE (80.2)
HousingQA	Llama 3.1 8B	55.4	62.3	59.1	59.0	—	-3.3	Raw question RAG (62.3)
HousingQA	Llama 3.3 70B	44.8	62.1	62.2	59.6	—	-2.5	HyDE (62.2)

Table 6. Gold-passage oracle controls. Accuracy values are percentages; deltas are percentage points over included row pairs.

Dataset	Model	Raw question RAG	Snap-HyRE	Gold	Gold vs. raw pp
BarExamQA	Llama 3.1 8B	54.5	56.9	60.0	+5.5
BarExamQA	Gemma 4 26B	78.0	82.0	78.6	+0.6
BarExamQA	Llama 3.3 70B	74.6	79.7	79.2	+4.6
HousingQA	Llama 3.1 8B	62.3	59.0	64.3	+2.0
HousingQA	Llama 3.3 70B	62.1	59.6	67.3	+5.2

Table 7. Neighbor-dilution diagnostic for oracle rows. Positive deltas mean nearby retrieved text helped beyond the gold passage alone.

Dataset	Model	Gold	Gold + neighbors	Δ pp
BarExamQA	Llama 3.1 8B	60.0	61.8	+1.8
BarExamQA	Gemma 4 26B	78.6	80.7	+2.1
BarExamQA	Llama 3.3 70B	79.2	77.8	-1.4

Table 8. Answer-stage token accounting for rows with token logs. First-stage query-generation usage is excluded. All methods use one scored answer call per question except Rewrite, which uses two.

Method	Cells	Input tok./q	Output tok./q	Correct / 1M tok.
LLM	5	131	237	1487.2
Raw question RAG	5	2295	339	241.5
HyDE	5	2237	340	244.5
Snap-HyRE (ours)	4	2001	376	268.3
Gold	5	1447	215	404.5
Rewrite	3	1969	555	284.2

A.3. Top-k Retrieval Diagnostics

Tables 9 and 10 and Figure 4 show how retrieval exposure changes as the cutoff grows from one to ten passages.

Table 9. HousingQA raw retrieval with and without the jurisdiction filter. Values are percentages from complete top-10 caches.

Corpus scope	Hit@1	Hit@3	Hit@5	Hit@10	MRR@10
National corpus	0.8	1.9	2.8	5.1	1.8
Jurisdiction metadata filter	15.5	29.5	36.9	48.1	24.8

Table 10. Top- k retrieval diagnostics from complete top-10 caches. Values are percentages; mean rows average model-specific generated-query caches.

Dataset	Model/scope	Method	n	Hit@3	Hit@5	Hit@10	MRR@10
BarExamQA	Shared	Raw question RAG	1195	0.9	1.4	2.2	0.8
BarExamQA	Mean over 3 models	HyDE	1195	6.4	10.0	16.7	5.9
BarExamQA	Mean over 3 models	Snap-HyRE (ours)	1195	7.1	10.9	17.3	6.3
BarExamQA	Llama 3.1 8B	HyDE	1195	5.5	8.3	13.5	5.2
BarExamQA	Llama 3.1 8B	Snap-HyRE (ours)	1195	6.3	9.5	14.8	5.4
BarExamQA	Gemma 4 26B	HyDE	1195	7.1	11.4	19.1	6.4
BarExamQA	Gemma 4 26B	Snap-HyRE (ours)	1195	7.9	12.1	18.7	6.9
BarExamQA	Llama 3.3 70B	HyDE	1195	6.5	10.5	17.6	6.1
BarExamQA	Llama 3.3 70B	Snap-HyRE (ours)	1195	7.2	11.0	18.5	6.6
HousingQA	State-filtered shared	Raw question RAG	6853	29.5	36.9	48.1	24.8
HousingQA	Mean over 2 full models	HyDE	6853	24.7	30.9	38.5	20.7
HousingQA	Mean over 2 full models	Snap-HyRE (ours)	6853	20.7	26.3	34.1	17.6
HousingQA	State-filtered Llama 3.1 8B	HyDE	6853	21.6	27.0	34.5	17.9
HousingQA	State-filtered Llama 3.1 8B	Snap-HyRE (ours)	6853	23.4	29.6	37.0	19.9
HousingQA	State-filtered Llama 3.3 70B	HyDE	6853	27.9	34.9	42.6	23.6
HousingQA	State-filtered Llama 3.3 70B	Snap-HyRE (ours)	6853	18.1	23.1	31.3	15.3

BarExamQA generated passages raise the chance of seeing the labeled evidence throughout the top-10 list, although MRR remains modest because the exact supporting passage is rarely ranked first. HousingQA behaves differently: the jurisdiction filter is the major retrieval correction, and raw state-filtered questions already retrieve relevant evidence often.

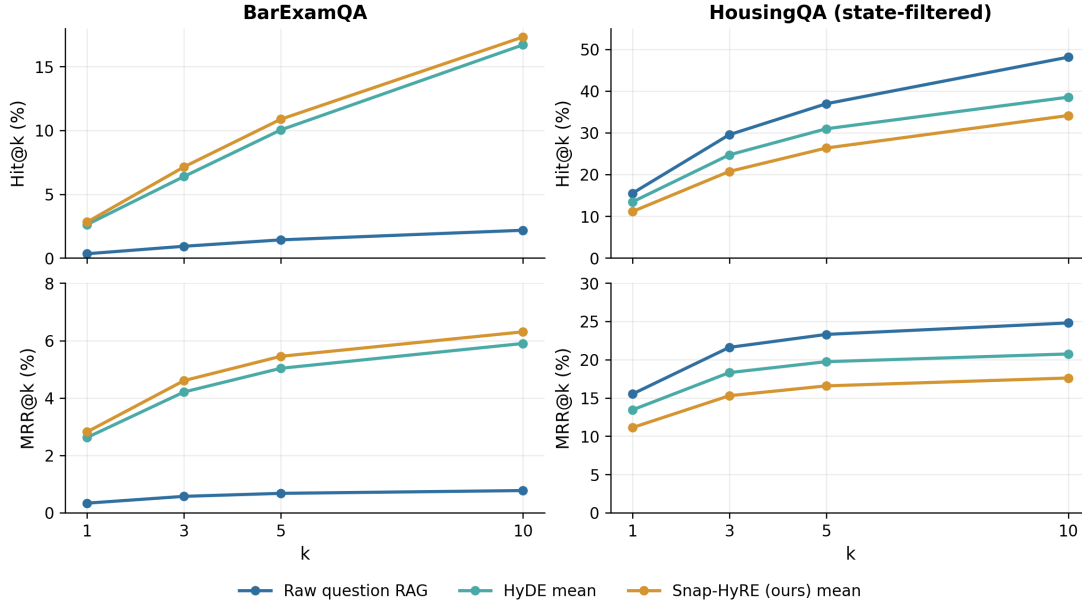


Figure 4. Top- k retrieval exposure and rank quality for full top-10 caches. BarExamQA generated-query curves average over Llama 3.1 8B, Gemma 4 26B, and Llama 3.3 70B; HousingQA averages over Llama 3.1 8B and Llama 3.3 70B with the jurisdiction state filter.

A.4. Statute/Passage Exemplar Probe

Table 11. Statute/passage exemplar lift, Gemma 4 26B q500 retrieval probes (Hit@5 / MRR@5, %). Δ is the exemplar lift over canonical Snap-HyRE. Reference raw-question Hit@5: BarExamQA 1.6, HousingQA (state-filtered) 36.9, Legal-Link-EU 90.0.

Dataset	Slice	Snap-HyRE	+ Exemplar	Δ Hit@5 / MRR@5
BarExamQA	q500	13.0 / 6.4	13.6 / 6.3	+0.6 / -0.1
HousingQA	q500 (state-filt.)	38.2 / 24.3	41.2 / 26.5	+3.0 / +2.2
Legal-Link-EU	q500	68.2 / 55.6	75.8 / 62.6	+7.6 / +7.0

The exemplar variant adds one fixed sanitized real-passage exemplar to the Snap-HyRE generation prompt — the exemplar’s own question, answer, and document id are stripped. The same q500 retrieval interfaces are used as the corresponding canonical Snap-HyRE rows: BarExamQA uses the shared support-passage collection; HousingQA uses the jurisdiction state filter; Legal-Link-EU uses the EUR-Lex evidence corpus with the same CROSS_ENCODER_MAX_CHARS=22000 setting as the main Snap-HyRE cache. The retrieval probes are computed directly from the cached passage-id lists; paired answer accuracy was scored only on the HousingQA q500 slice (62.8% exemplar vs 63.0% canonical Snap-HyRE, $-0.20pp$, $p = 1.0$).

A.5. Reproducibility Statement

All tables and figures are derived from the result records used for this draft. The released package will include the records, plotting code, and paper sources needed to reproduce the reported matrices and figures.